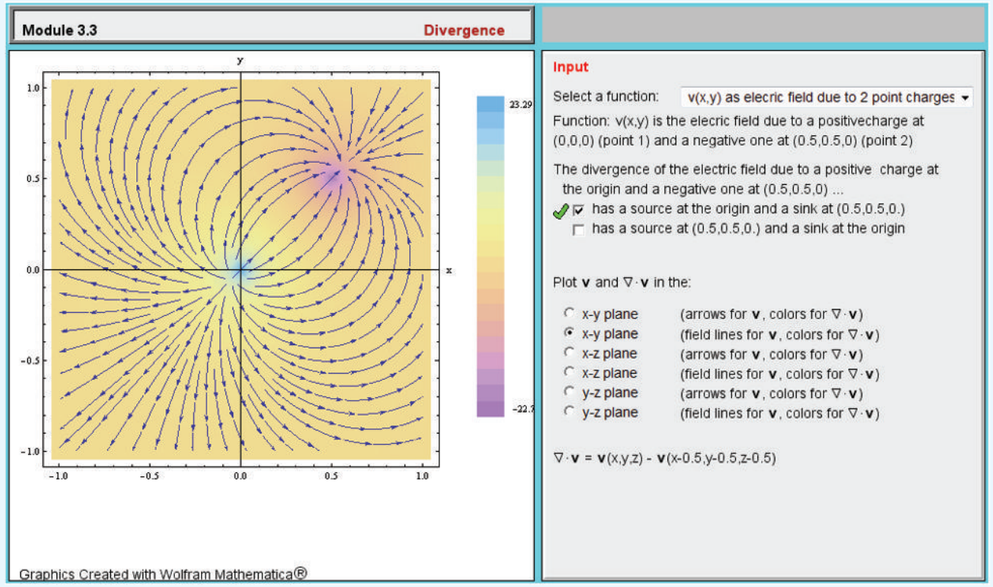
Module 3.3 Divergence Select a vector function $\mathbf{f}(x,y,z)$, evaluate its divergence, and display both in an appropriate 2-D plane.



(b) From the expression given in Appendix C for the divergence of a vector in spherical coordinates, it follows that

$$\begin{aligned}\nabla\cdot\mathbf{E} &= \frac{1}{R^2}\frac{\partial}{\partial R}(R^2E_R)+\frac{1}{R\sin\theta}\frac{\partial}{\partial\theta}(E_\theta\sin\theta)\\ &\quad+\frac{1}{R\sin\theta}\frac{\partial E_\phi}{\partial\phi}\\ &= \frac{1}{R^2}\frac{\partial}{\partial R}(a^3\cos\theta)+\frac{1}{R\sin\theta}\frac{\partial}{\partial\theta}\left(-\frac{a^3\sin^2\theta}{R^2}\right)\\ &= 0-\frac{2a^3\cos\theta}{R^3}=-\frac{2a^3\cos\theta}{R^3}.\end{aligned}$$

At $R=a/2$ and $\theta=0$, $\nabla\cdot\mathbf{E}\big|_{(a/2,0,\pi)}=-16$.

Exercise 3-13: Given $\mathbf{A}=e^{-2y}(\hat{\mathbf{x}}\sin 2x+\hat{\mathbf{y}}\cos 2x)$, find $\nabla\cdot\mathbf{A}$.

Answer: $\nabla\cdot\mathbf{A}=0$. (See EM.)

Exercise 3-14: Given $\mathbf{A}=\hat{\mathbf{r}}r\cos\phi+\hat{\boldsymbol{\phi}}r\sin\phi+\hat{\mathbf{z}}3z$, find $\nabla\cdot\mathbf{A}$ at $(2,0,3)$.

Answer: $\nabla\cdot\mathbf{A}=6$. (See EM.)

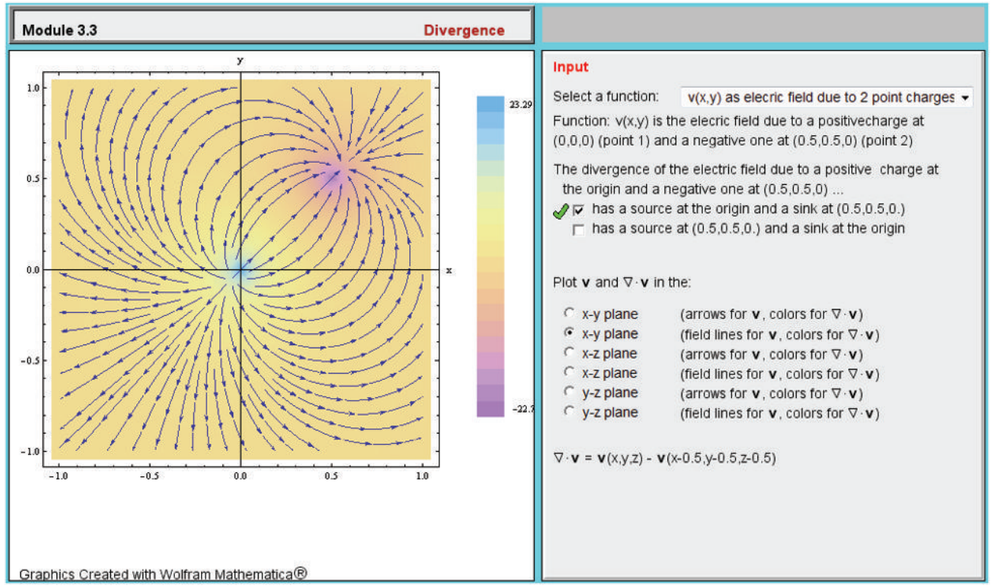
Exercise 3-15: If $\mathbf{E}=\hat{\mathbf{R}}AR$ in spherical coordinates, calculate the flux of $\mathbf{E}$ through a spherical surface of radius a, centered at the origin.

Answer: $\oint_S\mathbf{E}\cdot d\mathbf{s}=4\pi Aa^3$. (See EM.)

Exercise 3-16: Verify the divergence theorem by calculating the volume integral of the divergence of the field $\mathbf{E}$ of Exercise 3.15 over the volume bounded by the surface of radius a.

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模块 3.3 散度 选择矢量函数 f(x, y, z)，评估其散度，并将两者显示在适当的 2-D 平面中。



(b) 根据附录 C 给出的球坐标系向量散度表达式，可以得出：

$$\begin{aligned}\nabla\cdot\mathbf{E} &= \frac{1}{R^2}\frac{\partial}{\partial R}(R^2E_R)+\frac{1}{R\sin\theta}\frac{\partial}{\partial\theta}(E_\theta\sin\theta)\\ &\quad+\frac{1}{R\sin\theta}\frac{\partial E_\phi}{\partial\phi}\\ &= \frac{1}{R^2}\frac{\partial}{\partial R}(a^3\cos\theta)+\frac{1}{R\sin\theta}\frac{\partial}{\partial\theta}\left(-\frac{a^3\sin^2\theta}{R^2}\right)\\ &= 0-\frac{2a^3\cos\theta}{R^3}=-\frac{2a^3\cos\theta}{R^3}.\end{aligned}$$

在 R = a/2 和 θ = 0、$\nabla\cdot\mathrm{E}\big|_{(a/2,0,\pi)}=-16$ 处。

练习 3-13：给定 $\mathrm{A}=\mathrm{e}^{-2y}(\hat{\mathrm{x}}\sin 2\mathrm{x}+\hat{\mathrm{y}}\cos 2\mathrm{x})$，求 ∇· A。

答案：∇· A = 0。（参见EM。）

练习 3-14：给定 A = r̂ cos ϕ + ϕ̂r sin ϕ + ẑ3z，在 (2, 0, 3) 处找到 ∇· A。

Answer: $\nabla\cdot\mathbf{A}=6$. (See EM.)

练习3-15：如果球坐标中的E = R̂AR，计算E穿过以原点为中心、半径为a的球面的通量。

Answer: $\oint_S\mathbf{E}\cdot d\mathbf{s}=4\pi Aa^3$. (See EM.)

练习3-16：通过计算练习3.15中E场的散度在半径a的表面所包围的体积上的体积积分来验证散度定理。

Exercise 3-17: The arrow representation in **Fig. E3.17** represents the vector field $\mathbf{A} = \hat{\mathbf{x}}x - \hat{\mathbf{y}}y$. At a given point in space, $\mathbf{A}$ has a positive divergence $\nabla \cdot \mathbf{A}$ if the net flux flowing outward through the surface of an imaginary infinitesimal volume centered at that point is positive, $\nabla \cdot \mathbf{A}$ is negative if the net flux is into the volume, and $\nabla \cdot \mathbf{A} = 0$ if the same amount of flux enters into the volume as leaves it. Determine $\nabla \cdot \mathbf{A}$ everywhere in the x–y plane.

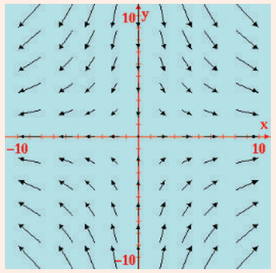


Figure E3.17

Answer: $\nabla \cdot \mathbf{A} = 0$ everywhere. (See EM.)

3-6 Curl of a Vector Field

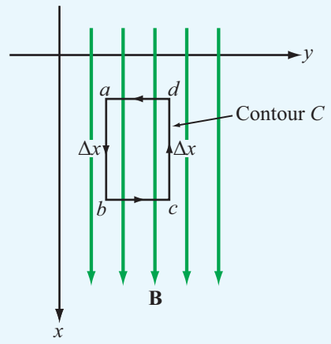
So far we have defined and discussed two of the three fundamental operators used in vector analysis: the gradient of a scalar and the divergence of a vector. Now we introduce the ***curl operator***. The curl of a vector field $\mathbf{B}$ describes its rotational property, or ***circulation***. The circulation of $\mathbf{B}$ is defined as the line integral of $\mathbf{B}$ around a closed contour C;

$$\text{Circulation} = \oint_C \mathbf{B} \cdot d\mathbf{l}. \tag{3.99}$$

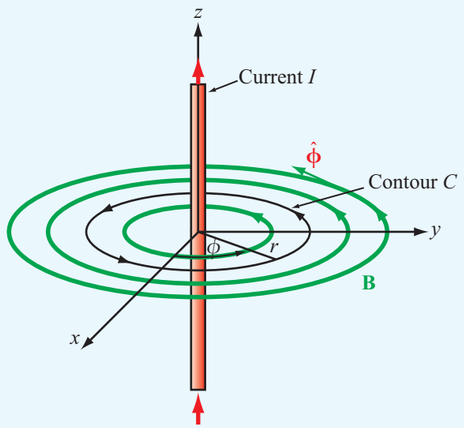
To gain a physical understanding of this definition, we consider two examples. The first is for a uniform field $\mathbf{B} = \hat{\mathbf{x}}B_0$, whose field lines are as depicted in **Fig. 3-23(a)**. For the rectangular contour *abcd* shown in the figure, we have

$$\begin{aligned}\text{Circulation} &= \int_a^b \hat{\mathbf{x}}B_0 \cdot \hat{\mathbf{x}}\, dx + \int_b^c \hat{\mathbf{x}}B_0 \cdot \hat{\mathbf{y}}\, dy \\ &\quad + \int_c^d \hat{\mathbf{x}}B_0 \cdot \hat{\mathbf{x}}\, dx + \int_d^a \hat{\mathbf{x}}B_0 \cdot \hat{\mathbf{y}}\, dy \\ &= B_0\,\Delta x - B_0\,\Delta x = 0,\end{aligned} \tag{3.100}$$

where $\Delta x = b - a = c - d$ and, because $\hat{\mathbf{x}} \cdot \hat{\mathbf{y}} = 0$, the second and fourth integrals are zero. According to Eq. (3.100), *the circulation of a uniform field is zero.*



(a) Uniform field



(b) Azimuthal field

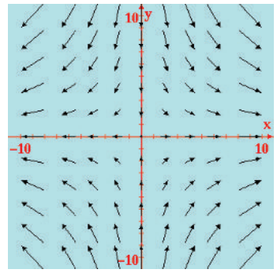
Figure 3-23 Circulation is zero for the uniform field in (a), but it is not zero for the azimuthal field in (b).

Next, we consider the magnetic flux density $\mathbf{B}$ induced by an infinite wire carrying a dc current I. If the current is in free space and it is oriented along the z direction, then from Eq. (1.13),

$$\mathbf{B} = \hat{\boldsymbol{\phi}}\frac{\mu_0 I}{2\pi r}, \tag{3.101}$$

where μ_0 is the permeability of free space and r is the radial distance from the current in the x–y plane. The direction of $\mathbf{B}$ is along the azimuth unit vector $\hat{\boldsymbol{\phi}}$. The field lines of $\mathbf{B}$ are concentric circles around the current, as shown in **Fig. 3-23(b)**.

练习3-17：图E3.17中的箭头表示向量场A = x̂ x − ŷ y 。在空间中的给定点，如果通过以该点为中心的虚无限小体积表面向外流动的净通量为正，则 A 具有正散度 ∇ · A；如果净通量流入该体积，则 ∇ · A 为负；如果进入该体积的通量与离开该体积的通量相同，则 ∇·A = 0。确定 x−y 平面上任意位置的 ∇ · A。



图E3.17

答案：∇ · A = 0 到处都是。 （参见EM。）

3-6 矢量场的旋度

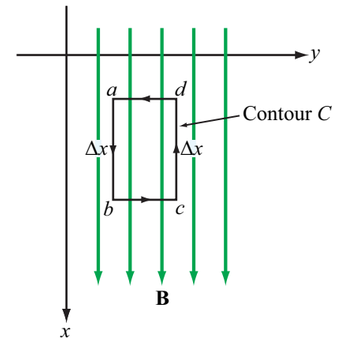
到目前为止，我们已经定义并讨论了向量分析中使用的三个基本运算符中的两个：标量的梯度和向量的散度。现在我们介绍curl运算符。矢量场 B 的旋度描述了它的旋转属性或循环。 B的环流定义为B绕封闭轮廓C的线积分；

$$\text{Circulation} = \oint_C \mathbf{B} \cdot d\mathbf{l}. \tag{3.99}$$

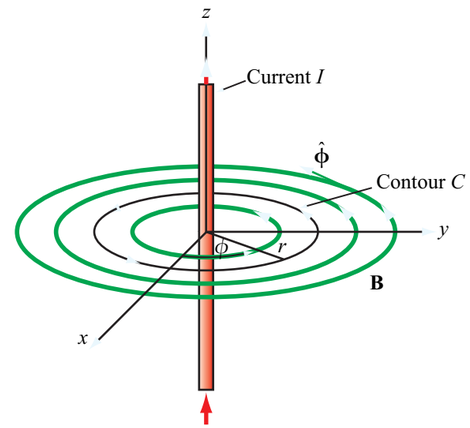
为了获得对这个定义的物理理解，我们考虑两个例子。第一个是均匀场 B = x̂B₀，其场线如图 3-23(a) 所示。对于图中所示的矩形轮廓 abcd，我们有

$$\begin{aligned}\text{Circulation} &= \int_a^b \hat{\mathbf{x}}B_0 \cdot \hat{\mathbf{x}}\, dx + \int_b^c \hat{\mathbf{x}}B_0 \cdot \hat{\mathbf{y}}\, dy \\ &\quad + \int_c^d \hat{\mathbf{x}}B_0 \cdot \hat{\mathbf{x}}\, dx + \int_d^a \hat{\mathbf{x}}B_0 \cdot \hat{\mathbf{y}}\, dy \\ &= B_0\,\Delta x - B_0\,\Delta x = 0,\end{aligned} \tag{3.100}$$

其中 Δx = b − a = c − d 并且，由于 x̂ · ŷ = 0，第二个和第四个积分为零。根据方程。 （3.100），均匀场的环流为零。



(a) Uniform field



(b) Azimuthal field

图 3-23 (a) 中均匀场的环流为零，但 (b) 中方位角场的环流不为零。

接下来，我们考虑由承载直流电流 I 的无限导线感应的磁通密度 B。如果电流位于自由空间中并且沿 z 方向定向，则根据等式： (1.13),

$$\mathbf{B} = \hat{\boldsymbol{\phi}}\frac{\mu_0 I}{2\pi r}, \tag{3.101}$$

其中 μ₀ 是自由空间的磁导率，r 是 x-y 平面中距电流的径向距离。 B的方向沿着方位角单位向量$\hat{\boldsymbol{\phi}}$。 B的磁力线是围绕电流的同心圆，如图3-23(b)所示。

For a circular contour C of radius r centered at the origin in the x–y plane, the differential length vector $d\mathbf{l} = \hat{\boldsymbol{\phi}} r\, d\phi$, and the circulation of $\mathbf{B}$ is

$$\text{Circulation} = \oint_C \mathbf{B}\cdot d\mathbf{l} = \int_0^{2\pi} \hat{\boldsymbol{\phi}}\frac{\mu_0 I}{2\pi r}\cdot\hat{\boldsymbol{\phi}} r\, d\phi = \mu_0 I. \quad (3.102)$$

In this case, the circulation is not zero. However, had the contour C been in the x–z or y–z planes, $d\mathbf{l}$ would not have had a $\hat{\boldsymbol{\phi}}$ component, and the integral would have yielded a zero circulation. Clearly, the circulation of $\mathbf{B}$ depends on the choice of contour and the direction in which it is traversed. To describe the circulation of a tornado, for example, we would like to choose our contour such that the circulation of the wind field is maximum, and we would like the circulation to have both a magnitude and a direction with the direction being toward the tornado's vortex. The curl operator embodies these properties. The curl of a vector field $\mathbf{B}$, denoted curl $\mathbf{B}$ or $\nabla\times\mathbf{B}$, is defined as

$$\nabla\times\mathbf{B} = \text{curl}\,\mathbf{B} = \lim_{\Delta s\to 0}\frac{1}{\Delta s}\left[\hat{\mathbf{n}}\oint_C \mathbf{B}\cdot d\mathbf{l}\right]_{\max}. \quad (3.103)$$

► Curl $\mathbf{B}$ is the circulation of $\mathbf{B}$ per unit area, with the area Δs of the contour C being oriented such that the circulation is maximum. ◄

The direction of curl $\mathbf{B}$ is $\hat{\mathbf{n}}$, the unit normal of Δs, defined according to the right-hand rule with the four fingers of the right hand following the contour direction $d\mathbf{l}$ and the thumb pointing along $\hat{\mathbf{n}}$ (Fig. 3-24). When we use the notation $\nabla\times\mathbf{B}$ to denote curl $\mathbf{B}$, it should *not* be interpreted as the cross product of ∇ and $\mathbf{B}$.

For a vector $\mathbf{B}$ specified in Cartesian coordinates as

$$\mathbf{B} = \hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z, \quad (3.104)$$

it can be shown, through a rather lengthy derivation, that Eq. (3.103) leads to

$$\nabla\times\mathbf{B} = \hat{\mathbf{x}}\left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z}\right) + \hat{\mathbf{y}}\left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x}\right) + \hat{\mathbf{z}}\left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y}\right) = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ B_x & B_y & B_z \end{vmatrix}. \quad (3.105)$$

Expressions for $\nabla\times\mathbf{B}$ are given in Appendix C for the three orthogonal coordinate systems considered in this chapter.

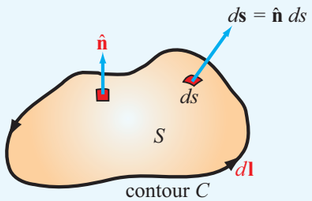


Figure 3-24 The direction of the unit vector $\hat{\mathbf{n}}$ is along the thumb when the other four fingers of the right hand follow $d\mathbf{l}$.

3-6.1 Vector Identities Involving the Curl

For any two vectors $\mathbf{A}$ and $\mathbf{B}$ and scalar V,

$$(1)\quad \nabla\times(\mathbf{A}+\mathbf{B}) = \nabla\times\mathbf{A} + \nabla\times\mathbf{B}, \quad (3.106a)$$

$$(2)\quad \nabla\cdot(\nabla\times\mathbf{A}) = 0, \quad (3.106b)$$

$$(3)\quad \nabla\times(\nabla V) = 0. \quad (3.106c)$$

3-6.2 Stokes's Theorem

► ***Stokes's theorem*** converts the surface integral of the curl of a vector over an open surface S into a line integral of the vector along the contour C bounding the surface S. ◄

For the geometry shown in Fig. 3-24, ***Stokes's theorem*** states

$$\int_S (\nabla\times\mathbf{B})\cdot d\mathbf{s} = \oint_C \mathbf{B}\cdot d\mathbf{l}. \quad (3.107)$$

(Stokes's theorem)

Its validity follows from the definition of $\nabla\times\mathbf{B}$ given by Eq. (3.103). If $\nabla\times\mathbf{B} = 0$, the field $\mathbf{B}$ is said to be ***conservative*** or ***irrotational*** because its circulation, represented by the right-hand side of Eq. (3.107), is zero, irrespective of the contour chosen.

对于 x-y 平面上以原点为圆心、半径为 r 的圆形轮廓 C，微分长度向量 dl = $\hat{\boldsymbol{\phi}}$r dϕ，B 的循环为

$$\text{Circulation} = \oint_C \mathbf{B}\cdot d\mathbf{l} = \int_0^{2\pi} \hat{\boldsymbol{\phi}}\frac{\mu_0 I}{2\pi r}\cdot\hat{\boldsymbol{\phi}} r\, d\phi = \mu_0 I. \quad (3.102)$$

在这种情况下，流通量不为零。然而，如果轮廓 C 位于 x -z 或 y-z 平面上，则 dl 不会
有一个$\hat{\boldsymbol{\phi}}$ 分量，积分将产生零循环。显然，B 的循环取决于轮廓的选择及其遍历的方向。例如，为了描述龙卷风的环流，我们希望选择轮廓以使风场的环流最大，并且我们希望环流具有大小和方向，方向朝向龙卷风的涡旋。 curl 运算符体现了这些属性。矢量场 B 的旋度，表示为旋度 B 或 ∇ × B，定义为

$$\nabla\times\mathbf{B} = \text{curl}\,\mathbf{B} = \lim_{\Delta s\to 0}\frac{1}{\Delta s}\left[\hat{\mathbf{n}}\oint_C \mathbf{B}\cdot d\mathbf{l}\right]_{\max}. \quad (3.103)$$

► 旋度B是每单位面积的B的循环，其中轮廓C的面积Δs被定向为使得循环最大。 ◄

卷曲 B 的方向为 n̂，即 Δs 的单位法线，根据右手定则定义，右手四指沿着轮廓方向 dl，拇指指向 n̂（图 3-24）。当我们使用符号 ∇ × B 来表示旋度 B 时，它不应该被解释为 ∇ 和 B 的叉积。

对于在笛卡尔坐标中指定的向量 B

$$\mathbf{B} = \hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z, \quad (3.104)$$

通过相当长的推导可以证明，方程。 (3.103) 导致

$$\nabla\times\mathbf{B} = \hat{\mathbf{x}}\left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z}\right) + \hat{\mathbf{y}}\left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x}\right) + \hat{\mathbf{z}}\left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y}\right) = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ B_x & B_y & B_z \end{vmatrix}. \quad (3.105)$$

附录 C 中给出了本章考虑的三个正交坐标系的 ∇ × B 的表达式。

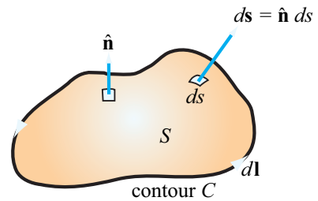


图3-24 当右手的其他四个手指跟随dl时，单位向量n̂的方向是沿着拇指的。

3-6.1 涉及旋度的向量恒等式

对于任意两个向量 A 和 B 以及标量 V，

$$(1)\quad \nabla\times(\mathbf{A}+\mathbf{B}) = \nabla\times\mathbf{A} + \nabla\times\mathbf{B}, \quad (3.106a)$$

$$(2)\quad \nabla\cdot(\nabla\times\mathbf{A}) = 0, \quad (3.106b)$$

$$(3)\quad \nabla\times(\nabla V) = 0. \quad (3.106c)$$

3-6.2 斯托克斯定理

► 斯托克斯定理将向量在开放曲面 S 上的旋度的曲面积分转换为向量沿包围曲面 S 的轮廓 C 的线积分。 ◄

对于图 3-24 所示的几何形状，斯托克斯定理指出

$$\int_S (\nabla\times\mathbf{B})\cdot d\mathbf{s} = \oint_C \mathbf{B}\cdot d\mathbf{l}. \quad (3.107)$$

(Stokes's theorem)

它的有效性来自等式给出的 ∇ × B 的定义。 (3.103)。如果 ∇ × B = 0，则称域 B 为保守域或无旋域，因为它的循环由等式右侧表示。 (3.107) 为零，无论选择什么轮廓。

Technology Brief 6: X-Ray Computed Tomography

► The word ***tomography*** is derived from the Greek words *tome*, meaning section or slice, and *graphia*, meaning writing. ◄

Computed tomography, also known as ***CT scan*** or ***CAT scan*** (for computed axial tomography), refers to a technique capable of generating 3-D images of the X-ray attenuation (absorption) properties of an object. This is in contrast to the traditional, X-ray technique that produces only a 2-D profile of the object (**Fig. TF6-1**). CT was invented in 1972 by British electrical engineer ***Godfrey Hounsfeld*** and independently by ***Allan Cormack***, a South African-born American physicist. The two inventors shared the 1979 Nobel Prize in Physiology or Medicine. Among diagnostic imaging techniques, CT has the decided advantage in having the sensitivity to image body parts on a wide range of densities—from soft tissue to blood vessels and bones.

Principle of Operation

In the system shown in **Fig. TF6-2**, the X-ray source and detector array are contained inside a circular structure through which the patient is moved along a conveyor belt. A CAT scan technician can monitor the reconstructed images to ensure that they do not contain artifacts such as streaks or blurry sections caused by movement on the part of the patient during the measurement process.

A CT scanner uses an ***X-ray source*** with a narrow slit that generates a ***fan-beam*** that is wide enough to encompass the extent of the body but only a few millimeters in thickness (**Fig. TF6-3**(a)). Instead of recording the attenuated X-ray beam on film, it is captured by an array of some 700 ***detectors***. The X-ray source and the detector array are mounted on a circular frame that rotates in steps of a fraction of a degree over a full 360° circle around the patient, each time recording an X-ray attenuation profile from a different angular perspective. Typically, 1,000 such profiles are recorded per each thin traverse slice of anatomy. In today's technology, this process is completed in less than 1 second. To image an entire part of the body, such as the chest or head, the process is repeated over multiple slices (layers), which typically takes about 10 seconds to complete.



Figure TF6-1 2-D X-ray image.

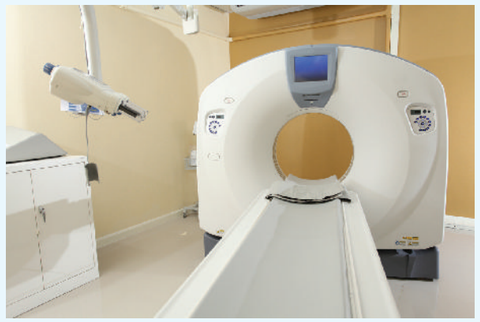


Figure TF6-2 CT scanner.

技术简介 6：X 射线计算机断层扫描

► 断层扫描一词源自希腊语 tome，意思是截面或切片，而 graphia，意思是书写。◄

计算机断层扫描，也称为 CT 扫描或 CAT 扫描（计算机轴向断层扫描），是指能够生成物体 X 射线衰减（吸收）特性的 3D 图像的技术。这与传统的 X 射线技术形成鲜明对比，传统的 X 射线技术仅产生物体的二维轮廓（图 TF6-1）。 CT 由英国电气工程师 Godfrey Hounsfeld 于 1972 年发明，并由南非出生的美国物理学家 Allan Cormack 独立发明。两位发明者共同获得 1979 年诺贝尔生理学奖或药品。在诊断成像技术中，CT 具有明显的优势，它对各种密度的身体部位（从软组织到血管和骨骼）的图像具有敏感性。

工作原理

在图 TF6-2 所示的系统中，X 射线源和探测器阵列包含在圆形结构内，患者沿着传送带移动通过该圆形结构。 CAT 扫描技术人员可以监控重建图像，以确保它们不包含测量过程中因患者运动而造成的条纹或模糊部分等伪影。

CT扫描仪使用带有窄缝的X射线源，产生扇形光束，其宽度足以包围整个身体，但厚度只有几毫米（图TF6-3（a））。衰减的 X 射线束不是在胶片上记录，而是由大约 700 个探测器组成的阵列捕获。 X 射线源和探测器阵列安装在一个圆形框架上，该框架以零点几度的步长在围绕患者的完整 360° 圆上旋转，每次从不同的角度记录 X 射线衰减剖面。通常，每个薄横切解剖切片都会记录 1,000 个这样的剖面。在当今的技术中，这个过程不到 1 秒即可完成。为了对身体的整个部位（例如胸部或头部）进行成像，需要在多个切片（层）上重复该过程，通常需要大约 10 秒才能完成。



图 TF6-1 二维 X 射线图像。

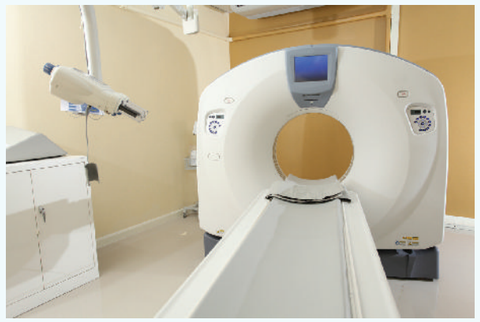
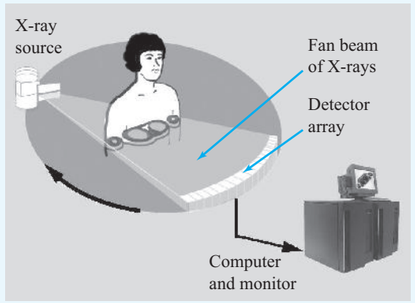
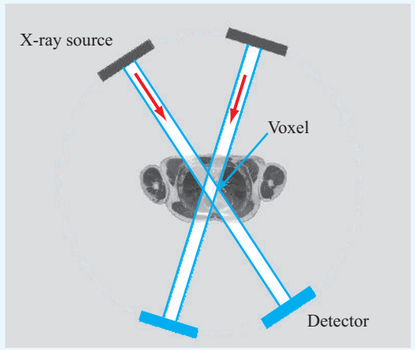


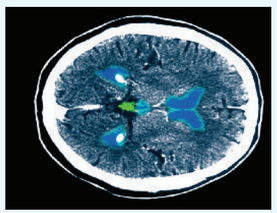
图 TF6-2 CT 扫描仪。



(a) CT scanner



(b) Detector measures integrated attenuation along anatomical path



(c) CT image of a normal brain

Figure TF6-3 Basic elements of a CT scanner.

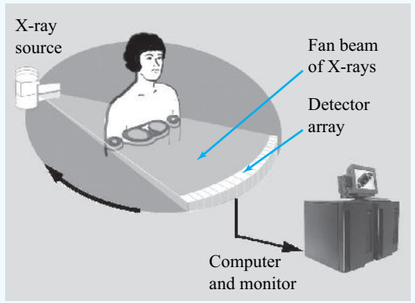
Image Reconstruction

For each anatomical slice, the CT scanner generates on the order of 7×10^5 measurements (1,000 angular orientations × 700 detector channels). Each measurement represents the integrated path attenuation for the narrow beam between the X-ray source and the detector (**Fig. TF6-3(b)**), and each volume element (***voxel***) contributes to 1,000 such measurement beams.

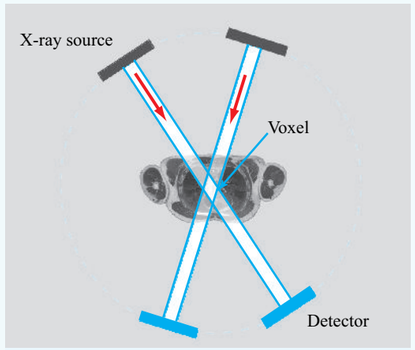
► Commercial CT machines use a technique called ***filtered back-projection*** to "reconstruct" an image of the attenuation rate of each voxel in the anatomical slice and, by extension, for each voxel in the entire body organ. This is accomplished through the application of a sophisticated matrix inversion process. ◄

A sample CT image of the brain is shown in **Fig. TF6-3(c)**.

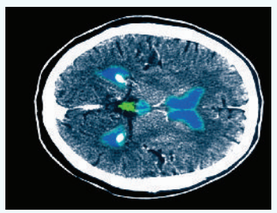
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(a) CT scanner



(b) Detector measures integrated attenuation along anatomical path



(c) 正常大脑的 CT 图像

图 TF6-3 CT 扫描仪的基本元件。

图像重建

对于每个解剖切片，CT 扫描仪生成大约 7×10^5 测量值（1,000 个角度方向 × 700 个探测器通道）。每个测量值代表X射线源和探测器之间窄光束的积分路径衰减（图TF 6-3（b））， 每个体积元素（体素）贡献1,000个这样的测量光束。

► 商用 CT 机使用一种称为滤波反投影的技术来“重建”解剖切片中每个体素的衰减率图像，进而重建整个身体器官中每个体素的衰减率图像。这是通过应用复杂的矩阵求逆过程来实现的。 ◄

大脑 CT 图像样本如图 TF6-3(c) 所示。